



Comparison of different methodologies for rainfall–runoff modeling: machine learning vs conceptual approach

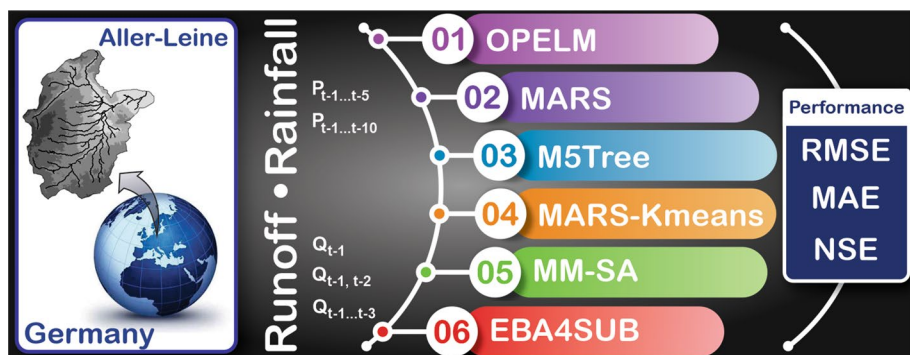
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Abstract

Accurate short-term rainfall–runoff prediction is essential for flood mitigation and safety of hydraulic structures and infrastructures. This study investigates the capability of four machine learning methods (MLM), optimal pruning extreme learning machine (OPELM), multivariate adaptive regression spline (MARS), M5 model tree (M5Tree, and hybridized MARS and Kmeans algorithm (MARS-Kmeans), in hourly rainfall–runoff modeling (considering 1-, 6- and 12-h horizons). Their results are compared with a conceptual method, Event-Based Approach for Small and Ungauged Basins (EBA4SUB) and multi-linear regression (MLR). Hourly rainfall and runoff data gathered from Ilme River watershed, Germany, were divided into two equal parts, and MLM were validated considering each part by swapping training and testing datasets. MLM were compared with EBA4SUB using four events and with respect to three statistics, root-mean-square errors (RMSE), mean absolute error (MAE) and Nash–Sutcliffe efficiency (NSE). Comparison results revealed that the newly developed hybridized MARS-Kmeans method performed superior to the OPELM, MARS, M5Tree and MLR methods in prediction of 1-, 6- and 12-h ahead runoff. Comparison with conceptual method showed that all the machine learning models outperformed the EBA4SUB and OPELM provided slightly better performance than the other three alternatives in event-based rainfall–runoff modeling.

Graphic abstract



Keywords Machine learning · Physically event-based conceptual method · EBA4SUB · Hourly rainfall–runoff modeling

1 Introduction

Rainfall–runoff modeling plays a vital role in the field of hydrology for flood disaster management, and for quantifying available water for drinking, agriculture purposes and hydropower scheduling. In recent decades, due to shortage of available water, accurate forecasting of runoff became a critical issue for hydrologists to preserve the available water resources (Wen et al. 2019). Rainfall–runoff modeling is a challenging task due to influence of numerous climatic factors such as air temperature, precipitation, wind speed, snow pack, solar radiation and humidity. In addition to climatic factors, the effect of catchment specific parameters (topography, shape, slope, altitude, soil type, land cover, moisture holding capacity) on runoff makes it more complicated and nonlinear process in nature (Santos et al. 2003; Feng et al. 2020). Long-term rainfall–runoff modeling (weekly, monthly) is important for dam construction, irrigation management and hydropower scheduling. However, short-term rainfall–runoff modeling (hourly or sub-hourly) is more required to save human lives and avoid infrastructures damage, in particular for small catchments characterized by a short response time of runoff to rainfall. In the past few decades, physically and conceptual-based models have been used extensively to model rainfall–runoff process. However, physical and conceptual-based models require much climatic input data that is very difficult to obtain in highly altitude scarce catchments (Zhu et al. 2019). Data intensive nature of these models makes them impractical especially in developing countries. In previous two decades, conventional time series models were used to investigate the rainfall–runoff process. However, due to influence of above-mentioned parameters, highly nonlinear and non-stationary behavior of rainfall–runoff process is difficult to be captured by conventional time series models. Therefore, in this decade, soft computing models have become very popular in many rainfall–runoff modeling studies, due to the data flexibility and ability to capture nonlinearity behavior (Wang et al. 2019). Artificial neural networks (ANN), support vector machines (SVM), extreme learning machines (ELM), M5 model tree (M5Tree) and multivariate adaptive regression splines (MARS) soft computing models have been utilized successfully in recent years (Fan et al. 2018; Fotovatikhah et al. 2018; Shamshirband et al. 2020; Fu et al. 2020; Taormina and Chau 2015; Gholami et al. 2016; Wu et al. 2013).

Due to more computation time and less ability to get global solution quickly of ANN and SVM soft computing models, ELM model is preferable and chosen in this study to model the rainfall–runoff process (Siqueira et al. 2012; Li and Cheng 2014; Yaseen et al. 2016a; Deo and Sahin 2016; Atiquzzaman and Kandasamy 2016; Nourani et al. 2017; Atiquzzaman and Kandasamy 2018; Li et al. 2019; Senent-Aparicio et al. 2019; Niu et al. 2019). Siqueira et al. (2012) utilized the ELM and echo state network (ESN) models for modeling seasonal runoff time series in comparison of multilayer perceptron (MLP) model. Results indicated the satisfactory performance of ELM model in comparison of other models. Atiquzzaman and Kandasamy (2018) evaluated the prediction accuracy of ELM model in comparison of genetic programming (GP) and SVM models for forecasting runoff data of two different catchments belongs to two different climate patterns. In results, they found that the ELM outperformed the other soft computing methods. Li et al. (2019) examined the ELM prediction capability in estimation of daily runoff, low runoff and peak runoff of a basin. They compared the results with random forest (RF) and SVM models and found that the ELM provided more precise results than the other soft computing models.

MARS model is selected in this study due to its short training process and ability to model complex nonlinear processes without strong model assumptions in comparison

of neural network models (Chou et al. 2004). Another selected method in this study is M5Tree model due to its easier large data handling and smaller computational cost in comparison of ANN and SVM models (Singh et.al 2010). Numerous successful applications of MARS and M5Tree models in the literature about runoff prediction also compelled authors to choose both models (Sharda et.al 2008; Adamowski et.al 2012; Sattari et.al 2013; Yaseen et al. 2016b; Adnan et al. 2018; Yin et al. 2018; Nourani et al. 2019; Al-Sudani et al. 2019; Mehdizadeh et al. 2019; Adnan et al. 2019a; Fathian et al. 2019). Sharda et.al (2008) evaluated the prediction accuracy of the MARS model to forecast runoff of three mid-Himalayan micro-watersheds and found that the MARS models provided reasonable runoff prediction results. Adamowski et al. (2012) predicted the runoff of three hydroclimatic stations in Uttaranchal State of India using the MARS model denoised with wavelet method (WT-MARS). The authors found that the WT-MARS model performed better than the standalone ANN model. Sattari et.al (2013) evaluated the prediction accuracy of M5Tree model in modeling daily runoff of Sohu River in Turkey. They compared the M5Tree model in comparison of SVM and found that M5Tree provided better results especially in multi-step ahead prediction. Nourani et al. (2019) applied the hybrid wavelet M5Tree model for rainfall–runoff modeling of two catchments' monthly and daily time series, and they found that the wavelet M5Tree outperformed the ANN and standalone M5Tree models.

Al-Sudani et al. (2019) utilized hybrid MARS model integrated with differential evolution algorithm (MARS-DE) for predicting runoff of Tigris River, Iraq. They evaluated the proposed method in comparison of least square support vector machine (LSSVM) and standalone MARS model and found that proposed hybrid methods provided more accurate results than the other two models. Mehdizadeh et al. (2019) applied the hybrid MARS and multiple linear regression (MTLR) models with integration of fractionally auto-regressive integrated moving average (MARS-FARIMA and MLR-FARIMA) and self-exciting threshold auto-regressive (MARS-SETAR and MLR-SETAR) models for predicting monthly runoff of two river gauges located in the USA. They compared the designed methods with standalone MARS and MLR models and found that the hybrid MARS-FARIMA and MARS-SETAR outperformed the other models. Adnan et al. (2019a, b) applied the MARS model in forecasting one month ahead runoff of two gauging stations of Swat River basin in comparison of LSSVM and M5Tree models. Their study results demonstrated the dominance of MARS model over other soft computing models in predicting monthly runoff of both gauging stations of Swat River basin. Fathian et al. (2019) evaluated the prediction accuracy of MARS model in comparison of ANN and random forest (RF) models for modeling the monthly streamflow of Brantford and Galt stations on Grand River, Canada. In the results, they found that the MARS and RF provided more accurate results than the ANN model.

Although the MARS and ELM models provided promising results in previous studies for hydrological modelling and predicted one month ahead runoff accurately, however, both methods could not handle the uncertainties in the short-term time series hydrological data, especially hourly rainfall–runoff data very well. Therefore, robust and hybrid models are needed to accurately model the hourly runoff data. In this study, a more robust model version of ELM was selected, i.e., OP-ELM due to the pruning procedure of the neurons to overcome the weakness of ELM in random initialization of the internal hidden layer weights. Fewer successfully applications of optimal pruning ELM (OPELM) were found in the literature for only monthly runoff modeling (Rezaie-Balf and Kisi 2018; Adnan et.al 2019). However, OPELM model has not been evaluated in hourly rainfall–runoff prediction. Hence, the present study will be the first time application of OPELM model for hourly

rainfall–runoff modeling. The findings of abovementioned studies revealed that hybrid models provided more accurate results than standalone models and that they modeled the nonlinear phenomena precisely. Therefore, in this study, a new hybrid model using integration of MARS with Kmeans method (MARS-Kmeans) was developed to overcome the weakness of standalone MARS models for capturing uncertainties in the rainfall–runoff data accurately. In addition, the recently developed conceptual hydrological model named Event-Based Approach for Small and Ungauged Basins (EBA4SUB) is also examined for short-term rainfall–runoff modeling in this paper.

The objectives of this study are as follows: (1) developing hybrid model with integration of MARS model with Kmeans method (MARS-Kmeans) to improve the prediction performance of the standalone MARS model, (2) determining the prediction accuracy of proposed hybrid method against alternative soft computing methods: OPELM, M5Tree and standalone MARS model, (3) comparing the selected soft computing methods in one hour ahead runoff prediction capability vs. a conceptual event-based method, EBA4SUB. All selected models were evaluated using whole dataset by swapping the training and testing datasets to determine the representative dataset. Selected models were also tested in runoff prediction for multiple horizons, i.e., 6 h and 12 h ahead. To the best of our knowledge, there is no published study using MARS-Kmeans method for hourly rainfall–runoff modeling. In addition, in the present time, first time selected soft computing methods (MARS-Kmeans, OPELM, M5Tree and standalone MARS) were compared with EBA4SUB hydrological model for rainfall–runoff modeling. These points led to the development of this study.

2 Materials and methods

2.1 Case study description

The selected case study is the Ilme River watershed, located in Germany and shown in Fig. 1. The watershed is pertaining to the Aller-Leine River basin, located in the south-eastern part of Lower Saxony. The Aller-Leine River basin is characterized by a heterogeneous landscape with mountains in the southern part, mainly covered by forests, and flatlands in the northern part. The climate is characterized by high annual precipitation and frost in the winter season, while the soil is mainly characterized by sandy soils, porous

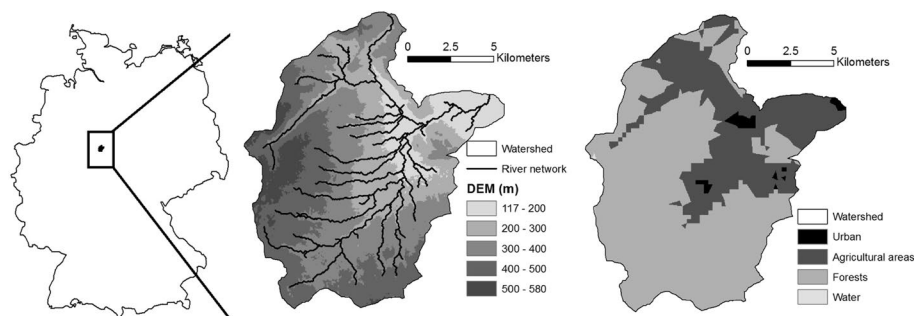


Fig. 1 Study area. Localization (left), DEM (center) and land cover (right)

quaternary aquifers and heath and moor vegetation. For a detailed description of the Aller-Leine River basin, refer to Wallner et al. (2012, 2013).

Regarding the Ilme River watershed, digital elevation model (DEM) at 30 m resolution and land cover data were retrieved from the ASTER GDEM (2009) and CORINE (2000) websites, respectively. The watershed is characterized by a contributing area of 148.4 km², elevations ranging from 117 to 580 m above sea level, a main channel length of 26.4 km and an average watershed slope of 16.8%. Regarding the land cover, it is characterized mainly by forests (66.5% of the total area), followed by agricultural areas (32% of the total area), urban areas (1.4% of the total area) and water bodies (0.1% of the total area). The rainfall and runoff data analyzed in the present work consist in a synchronized time series of hourly values of precipitation and discharge from January 1, 2004, to December 31, 2008, available thanks to the KOSTRA-DWD Program (2000).

2.2 The EBA4SUB model

The EBA4SUB (Event-Based Approach for Small and Ungauged Basins) conceptual rainfall–runoff model (Grimaldi and Petroselli 2015; Piscopia et al. 2015; Petroselli and Grimaldi 2018) is able to reconstruct the response of the basin to the rainfall, and it is based on an event-based approach that can be used in “small” and “ungauged” watersheds. For “small,” we mean that the watershed should have a total contributing area smaller than few hundreds of square kilometers, so that the assumption of spatially homogenous rainfall is still acceptable, while for “ungauged,” we mean that is lacking of discharge measurements.

EBA4SUB consists of three steps: (a) gross rainfall identification, (b) excess rainfall estimation and (c) runoff calculation. Input data of the model are gross rainfall information, DEM, land use and soil-type information. In the following, the three steps of the model are briefly recalled, while further details can be retrieved in the aforementioned references.

2.2.1 Gross rainfall identification

If depth–duration–frequency (DDF) curves are available, EBA4SUB can estimate the gross rainfall hyetograph selecting the desired rainfall duration and assuming a certain rainfall pattern (rectangular, triangular or Chicago). In this work, anyway, since we have observed rainfall data, we followed a different approach. First, we identified single independent rainfall events that are defined as rainfall sequences preceded and followed by at least 24 consecutive hours without rain, thus assuming the physical hypothesis that the initial abstraction phenomena return to the pre-event condition in one day (Grimaldi et al. 2012). Then, we modeled the rainfall–runoff relationship according to what is expressed in the following.

2.2.2 Excess rainfall estimation

Excess rainfall is estimated using the Curve Number for Green-Ampt (CN4GA) procedure (Grimaldi et al. 2013). CN4GA is a mixed approach combining the Natural Resources Conservation Service (NRCS) Curve Number (CN) procedure (NRCS 2008) with the Green-Ampt (GA) equation (Green and Ampt 1911). First, the NRCS-CN procedure is used to determine the ponding time and the cumulative excess rainfall, as follows:

$$P_n = \begin{cases} \frac{(P-Ia)^2}{P+S-Ia} & \text{for } P \geq Ia \\ 0 & \text{for } P < Ia \end{cases} \quad (1)$$

where P_n is the cumulative excess rainfall (mm), P the cumulative gross rainfall (mm), S the maximum potential watershed retention (mm), and Ia the initial abstraction due to the interception, infiltration and surface storage (mm). In the present work, Ia is assumed as the cumulative gross rainfall between the event start and the first moment when observed runoff is greater than zero, while S is derived from Eq. (1) computing cumulative rainfall and runoff from observed data. The second step of CN4GA aims in determining the temporal distribution of the excess rainfall employing the GA equation:

$$q_0(t) = \begin{cases} i(t) & \text{for } t < t_{\text{pond}} \\ K_s \left(1 + \frac{N_s}{F(t)} \right) & \text{for } t > t_{\text{pond}} \end{cases} \quad (2)$$

where $q_0(t)$ is the infiltration rate, t_{pond} the ponding time, $F(t)$ the cumulative infiltration, K_s the saturated hydraulic conductivity, and N_s the moisture-tension parameter. Equation (2) assumes that ponding time is reached when $P(t)$ equals to Ia , while K_s is automatically calibrated so that the total cumulative net runoff obtained by the NRCS-CN procedure equals to the corresponding one computed by the GA method. The other GA equation parameters are determined starting from the selected case study soil type based on look up tables.

2.2.3 Runoff calculation

The runoff consequent to excess rainfall is calculated using the width function-based instantaneous unit hydrograph (WFIUH), estimated based on case study DEM. First, the DEM is preprocessed: Pits and flat areas are removed, and the river network is extracted employing the threshold area criterion (here a 1 km^2 threshold value has been used). Then, the WFIUH is determined using the following formula (Grimaldi et al. 2012):

$$WFIUH(t) = \frac{L_c(x)}{v_c(x)} + \frac{L_h(x)}{v_h(x)} \quad (3)$$

where L_c and L_h are length of the path for channel and hillslope cell x of the DEM, respectively, and v_c and v_h are surface flow velocity for channel and hillslope cell, representing the parameters affecting the shape of the WFIUH. Finally, after the WFIUH is defined, the runoff $Q(t)$ is obtained according to:

$$Q(t) = A \int_0^t WFIUH(t - \tau) P_n(\tau) d\tau \quad (4)$$

where A is the basin area (km^2), t the rainfall duration (h), τ the time step, and $P_n(\tau)$ the excess rainfall (mm/h) determined by CN4GA. Regarding the determination of v_c and v_h , the hillslope surface flow velocity can be linked according to empirical formulas to local slope and land cover data (Grimaldi et al. 2012), while the channel kinematic component represents the only parameter to be determined. In the present work, its value can be calibrated for each analyzed event so that the modeled peak discharge is similar to the observed one.

2.3 OPELM

The OPELM methodology (Miche et al. 2010) is based on the original extreme learning machine algorithm (ELM), originally proposed by Huang et al. (2006), from which it borrows the original single-layer feedforward neural network (SLFN) construction. The OPELM algorithm is a more robust methodology regarding irrelevant variables situation. The main concept of the ELM is the random initialization weights of SLFN and biases. Then, the input parameters (weights/biases) do not required to be adjusted and it is possible to compute implicitly the hidden-layer output matrix and the output weights. The final network is obtained after very few steps and with very low computational cost. However, the ELM models tend to have problems when irrelevant or correlated variables are presented in the training dataset. Thus, the OPELM methodology was proposed, to perform a pruning of the irrelevant variables, via pruning of the related neurons of the SLFN built by the ELM, and it uses a combination of three different types of kernels (i.e., linear, sigmoid and Gaussian), for robustness and more generality, whereas the original ELM uses only sigmoid kernels.

2.4 M5Tree

M5 model tree (M5Tree) is a regression tree, which uses the same concept of a binary decision tree to split the data and uses linear regression functions at the nodes, instead of the class labels. The model is based on two steps: (i) create a decision tree using a splitting criterion and (ii) design the model tree by pruning the overgrown tree and replace the sub-trees with linear regression functions (Singh et al. 2010). Such splitting criterion is based on treating the *sd* of the class values that reach a leaf as a measure of error at that leaf. The expected reduction in this error is calculated as a result of testing each attribute at that leaf. The standard deviation reduction (*SDR*) is as follows:

$$SDR = sd(N) - \sum \frac{|N_i|}{|N|} sd(N_i) \quad (5)$$

where N is the examples that reach the leaf, N_i is the subset of examples with the i th outcome of the potential set, and *sd* is the standard deviation. The data in child nodes present less *sd* than those at the parent node, which makes it purer. Therefore, at the end, M5 selects the node that maximizes the expected error reduction, which may lead to a large tree-like structure inducing to an over-fitting. Then, step (ii) takes place when a sub-tree may be replaced by a leaf (Pal and Deswal 2009).

2.5 MARS

Friedman (1991) proposed the multivariate adaptive regression splines (MARS), which is a nonlinear and nonparametric method, and is able to predict a continuous dependent parameter in high-dimensional data through a forward–backward procedure. In order to initially fit the data, various basis functions are added to the model during the forward stepwise stage. Such basis functions are dependent on a spline function which split the dataset into piecewise linear segments. The MARS-based models are as follows:

$$f(x) = \delta_0 + \sum_{m=1}^M \delta_m h_m(X) \quad (6)$$

where δ_0 and δ_m are coefficients that can be computed by the minimum sum of squared errors from splines functions, $h_m(x)$ are spline functions, and M is the number of functions. There is a pruning state, which is a backward stage to eliminate the unnecessary basis functions through a generalized cross-validation criterion. Therefore, this process improves the prediction accuracy and M is determined (Kisi and Parmar 2016). More details about the MARS algorithm are available in Friedman (1991).

2.6 Hybridized MARS and Kmeans

There are several clustering techniques, and k -means clustering is one of them, which has been applied in different areas and is based on squared Euclidean distance (Hartigan and Wong 1979; Santos et al. 2017, 2019). The algorithm proceeds a partition of n observations into k clusters. Each observation is within a cluster with the nearest mean, resulting a partition of the data space into Voronoi cells (Milligan and Cooper 1985). The k -means algorithm iteratively minimizes the distances between each data point and its centroid to obtain the best solution for all the data points. Obtaining the minimal distances between the points means that data points were separated to form the most possible compact cluster with the least variance, i.e., no other iteration could have a lower average distance between the centroids and the data points found within them.

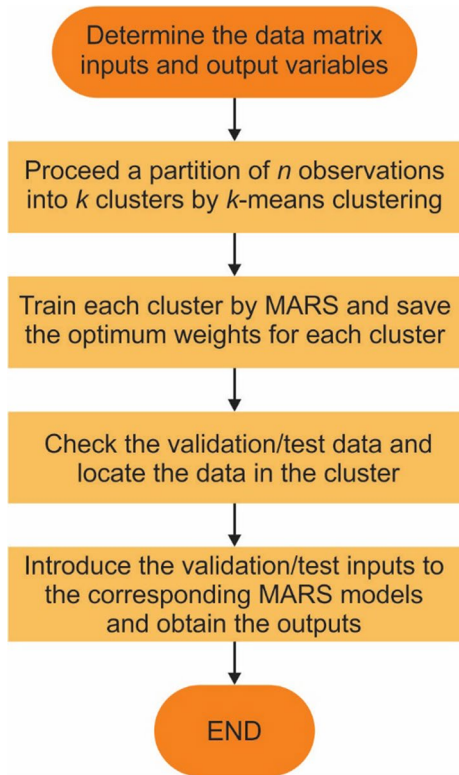
A MATLAB script was written to integrate the k -means clustering and MARS method, according to the following steps: (i) determine the data matrix of inputs and output variables; (ii) proceed partition of n observations into k clusters by k -means clustering; (iii) train each cluster by MARS and save the optimum weights or parameters for each cluster; (iv) check the validation/test data and locate the data in the cluster; and (v) introduce the validation/test inputs to the corresponding MARS models and obtain the outputs. The respective flowchart is shown in Fig. 2.

2.7 The assessment of the implemented methods

Four machine learning methods were applied in this study using different input combinations for hourly rainfall–runoff modelling adopting various input combinations. The implemented methods were: (i) optimally pruned extreme learning machine (OPELM), (ii) multivariate adaptive regression splines (MARS), (iii) MARS with *Kmeans clustering algorithm* (MARS-Kmeans) and (iv) M5 model tree (M5Tree). An ensemble multi-model, simple averaging called (MM-SA) was also implemented. The results of the machine learning methods were also compared with the event-based approach for small and ungauged basins (EBA4SUB). As a basis for comparison, models were assessed using three statistics indices: mean absolute error (MAE), root-mean-squared error (RMSE) and Nash–Sutcliffe efficiency (NSE), calculated as in the following:

$$MAE = \frac{1}{N} \sum_{i=1}^N |(Q_0)_i - (Q_C)_i| \quad (7)$$

Fig. 2 The working schema of MARS-Kmeans model



$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N [(Q_0)_i - (Q_C)_i]^2} \quad (8)$$

$$NSE = 1 - \frac{\sum_{i=1}^N [(Q_0)_i - (Q_C)_i]^2}{\sum_{i=1}^N [(Q_0)_i - \bar{Q}_0]^2}, \quad -\infty < NSE \leq 1 \quad (9)$$

where N is the number of data, and Q_o , Q_p , $\bar{Q}_0$ are the measured, calculated, mean measured, hourly runoff, respectively.

3 Results and discussion

This section presents and discusses the obtained results using the proposed machines learning models for the case study. Brief statistical characteristics of the rainfall and runoff data are reported in Table 1. Two distinct scenarios were considered in the presented study. For the first scenario (1st train-test), the models were developed using 50% of the dataset and validated using the remaining 50%. In the second scenario (2nd

Table 1 The statistical parameters of the applied data

	Whole data	Training	Test
<i>Precipitation, mm</i>			
Mean	0.113	0.107	0.119
Min	0	0	0
Max	21.78	21.78	14.25
Median	0	0	0
Skewness	12.14	14.97	9.88
Std dev	0.495	0.474	0.514
<i>Streamflow, m³/s</i>			
Mean	2.048	1.863	1.39
Min	0.30	0.30	0.43
Max	29.46	16.44	29.46
Median	1.33	1.25	1.39
Skewness	3.640	2.968	3.72
Std dev	2.207	1.861	2.49

traintest), the models were trained using the validation data and validated using the training data with respect to the same splitting ratio (50/50%). Training and testing datasets were exchanged without applying randomization. For the two scenarios, we compared five combinations of runoff (Q) and rainfall (P) as shown in Table 2. Statistical indices such as RMSE, MAE, NSE and the linear regression analysis (scatterplot) are used to quantify the performance of the proposed models. For the OPELM method, different activation functions, linear, sigmoid, Gaussian and combination of these, were tried and the optimal one was selected. For the MARS and M5Tree models, defaults parameters were applied. Figures 3 and 4 show comparisons of measured hourly runoff values against the simulated data for the first (1st train-test) and second (2nd train-test) scenarios using the four machines learning models. Table 3 shows the accuracies of the OPELM models according to the five input combinations. According to Table 3, using only previous rainfall data as input variables, *i.e.*, combination (i) and (ii), the RMSE and MAE are higher and the NSE value was lower than 8%, which reflect very low accuracies and the incapacity of the OPELM model to correctly predict the hourly runoff using only the rainfall data. This can be partly explained by the fact that, for a short time interval (hours), the runoff process was only partly governed by the rainfall. Beyond

Table 2 The input combinations considered for hourly streamflow prediction

Input combination	Variables
(i)	$P_{t-1}, P_{t-2}, \dots, P_{t-5}$
(ii)	$P_{t-1}, P_{t-2}, \dots, P_{t-10}$
(iii)	$P_{t-1}, P_{t-2}, \dots, P_{t-5}, Q_{t-1}$
(iv)	$P_{t-1}, P_{t-2}, \dots, P_{t-5}, Q_{t-1}, Q_{t-2}$
(v)	$P_{t-1}, P_{t-2}, \dots, P_{t-5}, Q_{t-1}, Q_{t-2}, Q_{t-3}$

In the table, P_{t-1} and Q_{t-1} indicate the precipitation and streamflow value at time $t-1$ (one previous hour). Among the P inputs, input combination (i) was considered, and therefore, after the 1st combination, streamflows were added to this combination

Fig. 3 The scatterplot comparison of observed and predicted runoff by machine learning methods—1st train-test set

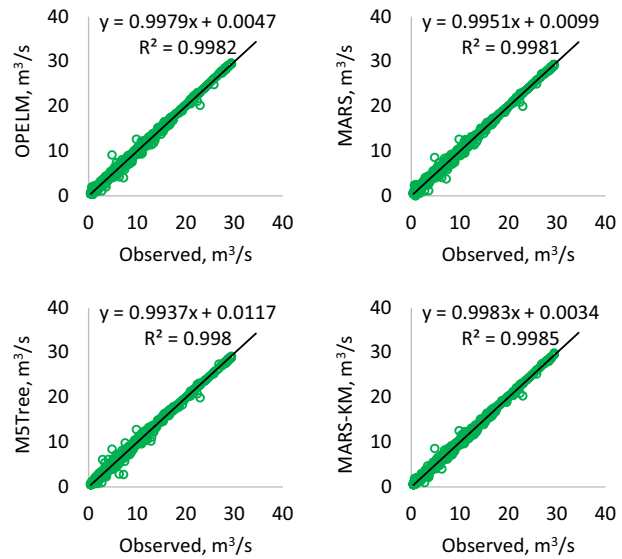
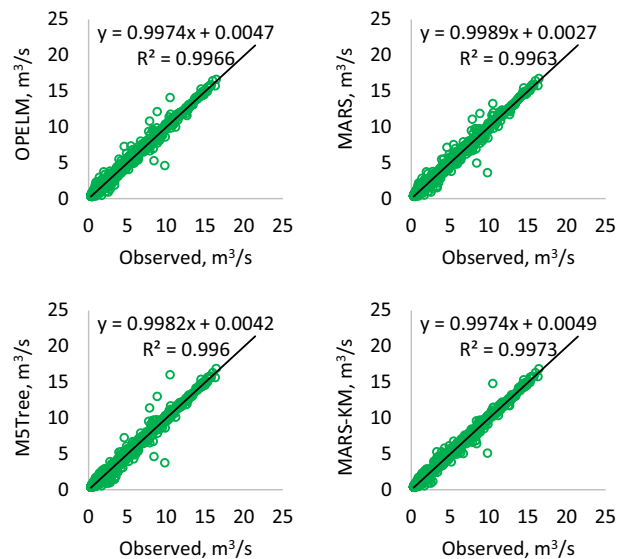


Fig. 4 The scatterplot comparison of observed and predicted runoff by machine learning methods—2nd train-test set



the second input combination (iii to v), for which previous runoff was included as input variables, the performances of the models were significantly improved, and we found a good correlation between the measured and calculated dataset, with lower RMSE and MAE, and high NSE values. The RMSE and MAE values for the third, fourth and fifth input combinations (Table 3) were lower than by the first and second input combination. While the accuracy for 1st train-test dataset was slightly improved compared to the 2nd train-test dataset, the accuracies were markedly improved with the third, fourth and fifth input combinations. The fourth input combination produced the best accuracy, regardless of RMSE, MAE and NSE values. Particularly, the OPELM model showed

Table 3 Test statistics of the OPELM using different combinations of streamflow and precipitation for hourly streamflow prediction

Statistics	Dataset	Input combination				
		(i)	(ii)	(iii)	(iv)	(v)
RMSE	1st train-test	2.45	2.4	0.123	0.106	0.143
	2nd train-test	1.86	1.84	0.117	0.109	0.111
	Mean	2.16	2.12	0.120	0.108	0.127
MAE	1st train-test	1.4	1.39	0.039	0.035	0.040
	2nd train-test	1.34	1.33	0.038	0.033	0.036
	Mean	1.37	1.36	0.039	0.034	0.038
NSE	1st train-test	0.032	0.071	0.998	0.998	0.997
	2nd train-test	0.002	0.024	0.996	0.997	0.997
	Mean	0.017	0.048	0.997	0.998	0.548

In the table, input combinations are (i) Pt-1, Pt-2, ..., Pt-5, (ii) Pt-1, Pt-2, ..., Pt-10, (iii) Pt-1, Pt-2, ..., Pt-5, Qt-1, (iv) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2 and (v) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2, Qt-3. Among the P inputs, input combination (i) was considered, and therefore, after the 1st combination, streamflows were added to this combination

remarkable improvement from the first ($\text{RMSE} = 2.45 \text{ m}^3/\text{s}$) to the fourth input combination ($\text{RMSE} = 0.106 \text{ m}^3/\text{s}$).

Table 4 shows the accuracies of the MARS model using the five input combinations. It is clear that the combination (iv) and (v) guaranteed the same performances with equally RMSE, MAE and NSE values, significantly higher than the first and second input combinations, and slightly higher than the third input combination. Specifically, the RMSE and MAE obtained with the combination (iv) ($\text{RMSE} = 0.108 \text{ m}^3/\text{s}$, $\text{MAE} = 0.032 \text{ m}^3/\text{s}$) were higher than (95.5% and 97.68%) those obtained with the second input combination ($\text{RMSE} = 2.4 \text{ m}^3/\text{s}$, $\text{MAE} = 1.38 \text{ m}^3/\text{s}$). Although the improvement between the third and

Table 4 Test statistics of the MARS using different combinations of streamflow and precipitation for hourly streamflow prediction

Statistics	Dataset	Input combination				
		(i)	(ii)	(iii)	(iv)	(v)
RMSE	1st train-test	2.44	2.4	0.13	0.108	0.108
	2nd train-test	1.87	1.85	0.122	0.113	0.113
	Mean	2.16	2.13	0.126	0.111	0.111
MAE	1st train-test	1.4	1.38	0.041	0.032	0.032
	2nd train-test	1.34	1.32	0.043	0.034	0.034
	Mean	1.37	1.35	0.042	0.033	0.033
NSE	1st train-test	0.039	0.076	0.997	0.998	0.998
	2nd train-test	−0.005	0.017	0.996	0.996	0.996
	Mean	0.017	0.047	0.997	0.997	0.997

In the table, input combinations are (i) Pt-1, Pt-2, ..., Pt-5, (ii) Pt-1, Pt-2, ..., Pt-10, (iii) Pt-1, Pt-2, ..., Pt-5, Qt-1, (iv) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2 and (v) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2, Qt-3. Among the P inputs, input combination (i) was considered, and therefore, after the 1st combination, streamflows were added to this combination

the fourth input combination is negligible taking into account the NSE value, the improvement in runoff forecasting was most obvious when the models were compared taking into account the RMSE for which we recorded a significant improvement: 16.92% and 7.37% for the 1st and 2nd dataset, respectively. The estimated RMSE, MAE and NSE values for the M5Tree model have strong variations as shown in Table 5. Overall, relationships between measured and calculated hourly runoff are very good for all input combinations beyond combination (iii) with limited scattering and slopes close to one (Figs. 3 and 4). The first and second input combinations, for which only the rainfall was used as input variable, have the lowest correlations to the measured data as shown by the poorest NSE value, and the RMSE and MAE were the highest. These results also suggest that using previous runoff data as input variable is appropriate and contribute to an improvement of the models' accuracy. From the third combination, the models' accuracy has progressed significantly, largely due to the addition of the runoff data, which can be improved further by the use of the fourth and five combinations. Mean RMSE, MAE and NSE for M5Tree model averaged across the five input combination show that, in general, combinations (iv) and (v) show a similar values (Table 5), except a slightly improvement in the RMSE in favor to the combination (v) and using the first (1st train-test) dataset, for which 6.61% improvement was achieved. In overall, the lower and negligible difference between the mean RMSE, MAE and NSE values between combinations (iv) and (v) also suggests an important finding; the low sensitivity of the model to the inclusion of the Q_{t-3} which may lead to conclude that, beyond the Q_{t-2} , the previous runoff data are indistinguishable by the model and the improvement become negligible: 0% improvement in terms of MAE and NSE values, and 3.33% in term of RMSE.

Table 6 lists the calculated RMSE, MAE and NSE values using the MARS-Kmeans. Among the five input combinations, there was considerable improvement in the prediction of the hourly streamflow compared to the previous discussed models (M5Tree, MARS and OPELM). For example, the accuracy of the first and second input combination (i and ii) for which only the previous rainfall data was included as input, was clearly improved, and the MARS-Kmeans model has demonstrated its ability to correctly and

Table 5 Test statistics of the M5Tree using different combinations of streamflow and precipitation for hourly streamflow prediction

Statistics	Dataset	Input combination				
		(i)	(ii)	(iii)	(iv)	(v)
RMSE	1st train-test	2.57	2.59	0.437	0.121	0.113
	2nd train-test	2.09	2.16	0.135	0.118	0.118
	Mean	2.33	2.38	0.286	0.120	0.116
MAE	1st train-test	1.47	1.49	0.055	0.032	0.030
	2nd train-test	1.38	1.39	0.034	0.030	0.030
	Mean	1.43	1.44	0.045	0.031	0.030
NSE	1st train-test	-0.068	-0.076	0.969	0.998	0.998
	2nd train-test	-0.258	-0.343	0.995	0.996	0.996
	Mean	-0.163	-0.210	0.982	0.997	0.997

In the table, input combinations are (i) Pt-1, Pt-2, ..., Pt-5, (ii) Pt-1, Pt-2, ..., Pt-10, (iii) Pt-1, Pt-2, ..., Pt-5, Qt-1, (iv) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2 and (v) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2, Qt-3. Among the P inputs, input combination (i) was considered, and therefore, after the 1st combination, streamflows were added to this combination

Table 6 Test statistics of the MARS-Kmeans using different combinations of streamflow and precipitation for hourly streamflow prediction

Statistics	Dataset	Input combination				
		(i)	(ii)	(iii)	(iv)	(v)
RMSE	1st train-test	1.15	1.12	0.122	0.098	0.094
	2nd train-test	0.83	1.14	0.105	0.099	0.097
	Mean	0.99	1.13	0.114	0.099	0.096
MAE	1st train-test	0.63	0.62	0.038	0.032	0.031
	2nd train-test	0.63	0.87	0.035	0.032	0.032
	Mean	0.63	0.75	0.037	0.032	0.032
NSE	1st train-test	0.788	0.798	0.998	0.998	0.999
	2nd train-test	0.803	0.625	0.997	0.997	0.997
	Mean	0.796	0.712	0.998	0.998	0.998

In the table, input combinations are (i) Pt-1, Pt-2, ..., Pt-5, (ii) Pt-1, Pt-2, ..., Pt-10, (iii) Pt-1, Pt-2, ..., Pt-5, Qt-1, (iv) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2 and (v) Pt-1, Pt-2, ..., Pt-5, Qt-1, Qt-2, Qt-3. Among the P inputs, input combination (i) was considered, and therefore, after the 1st combination, streamflows were added to this combination

accurately predict hourly runoff without recourse to the runoff data, while, as expected, the three other input combinations (*iii*, *iv* and *v*) showed no significant differences between the M5Tree, MARS, OPELM and the MARS-Kmeans models. To our knowledge, this is the first use of MARS-Kmeans approach for rainfall–runoff modelling, and compared to the standalone approaches, the results are very encouraging. For the first and second input combination (*i* and *ii*, Table 6) where only rainfall data were included in the model input, the mean RMSE and MAE values were 0.99 m³/s, 0.63 m³/s for the first input and 1.13 m³/s, 0.75 m³/s for the second input combination, respectively. Between the modelled and measured runoff, mean NSE was 0.796 for combination (*i*) and was 0.712 for combination (*ii*), with an improvement of 8.4%. In both combinations, there was a common tendency for combination (*i*) to be more accurate than combination (*ii*), which lead to conclude that inclusion of more previous rainfall data does not necessarily guaranty enough improvement in the model accuracy. Beyond the third combination, the models performed reasonably well in terms of the accuracy when considering the previous runoff as input variables (Table 6), and it is clear that the models become stable and the performances were relatively equal for the all three input combinations (*iii*, *iv* and *v*), and no obvious differences in overall model performance were observed between them. Most notably, MARS-Kmeans using combination (*iii*), (*iv*) and (*v*) produces NSE values in excesses of 0.99 and regression slopes close to 1 when compared to in situ measured hourly runoff (Figs. 3 and 4). For all three combinations, the mean RMSE and MAE were significantly the poorest, with the most obvious values obtained by the combination (*v*): (0.032 m³/s) and (0.096 m³/s), respectively. In almost every input combination, the addition of the previous runoff produced more accurate prediction of hourly runoff, and for some combinations who have excluded the previous runoff from the input variables, causes models performances to significantly degrade, as is the case with combination (*i*) and (*ii*). For numerical comparison, MARS-Kmeans with combination (*v*) decreases the RMSE and MAE of the same model using combination (*i*) by 90.30% and 94.2%, respectively, and increases the NSE value by 20.20%, respectively.

The second stage of comparison between the four machines learning models was focused on performance between models. Hourly runoff calculated using the MARS-Kmeans was compared to those calculated using OPELM, MARS and M5Tree models, for only the corresponding best input combination. Firstly, using the OPELM, MARS and M5Tree models, the calculated statistical indices identify the apparent poorly performances prediction of models using combination (i) and (ii), which produce a very low or negative NSE value using whether 1st or 2nd dataset (M5Tree, Table 5). Again, it is shown that increase in the input variables with only previous rainfall data from five for combination (i) to ten for combination (ii) does not always improve model performance; on the contrary, a large number of previous rainfall data produce a detrimental effect on model performances. This is particularly obvious in the MAE and RMSE calculated for M5Tree model, where the RMSE and MAE increase in approximately 2.10% and 0.6%, respectively. Nonetheless, it is important to note that significant improvement in terms of performance is achieved when previous runoff data were included in the input of the models, and this remark that needs to be made is more general for the all proposed machines learning models. Secondly, the exception was coming from the newly introduced MARS-Kmeans models, for which combination (i) and (ii), who have considered being very low for OPELM, MARS and M5Tree, performed reasonably well, and a significant improvement was achieved. Numerical comparison reveals that the MARS-Kmeans using combination (i) (Table 6) decreased the mean RMSE values of the OPELM, MARS and M5Tree by 54.16%, 54.16% and 57.51%, respectively (Table 3–5). In addition, the MARS-Kmeans using combination (i) achieved an improvement of 55.94%, 54.01% and 54.01%, in terms of MAE compared to the OPELM, MARS and M5Tree, respectively. According to Tables 3–6, the MARS-Kmeans using combination (v) was more accurate than the OPELM, MARS and M5Tree models, and it achieves 17.24%, 13.51% and 11.11% reduction in the RMSE values, respectively. It is also worth noting that taking into account the NSE values, the four machines learning models guarantied the same accuracy.

To further evaluate the general capability of the machines learning models, we have made a comparison between the models in predicting hourly runoff several hours in advance: (i) one hour ahead (Table 7), (ii) six hours ahead (Table 8) and (iii) twelve hours ahead (Table 9). The machine learning models were also compared with multi-linear

Table 7 Test statistics of the best OPELM, MARS, M5Tree, MARS-Kmeans, MM-SA and MLR in 1 h ahead streamflow prediction

Statistics	Dataset	OPELM	MARS	M5Tree	MARS-Kmeans	MM-SA	MLR
RMSE	1st train-test	0.106	0.108	0.113	0.094	0.105	0.108
	2nd train-test	0.109	0.113	0.118	0.097	0.109	0.112
	Mean	0.108	0.111	0.116	0.096	0.108	0.110
MAE	1st train-test	0.035	0.032	0.030	0.031	0.032	0.037
	2nd train-test	0.033	0.034	0.030	0.032	0.032	0.036
	Mean	0.034	0.033	0.030	0.032	0.032	0.037
NSE	1st train-test	0.998	0.998	0.998	0.999	0.998	0.998
	2nd train-test	0.997	0.996	0.996	0.997	0.997	0.996
	Mean	0.998	0.997	0.997	0.998	0.998	0.997

MM-SA is multi-model with simple averaging. The estimates of this model were simply calculated by averaging the estimates of the optimal OPELM, MARS, M5Tree and MARS-Kmeans models

Table 8 Test statistics of the best OPELM, MARS, M5Tree, MARS-Kmeans and MLR in 6 h ahead streamflow prediction

Statistics	Dataset	OPELM	MARS	M5Tree	MARS-Kmeans	MLR
RMSE	1st train-test	0.534	0.778	0.760	0.652	0.543
	2nd train-test	0.447	0.434	0.493	0.303	0.449
	Mean	0.491	0.606	0.627	0.478	0.496
MAE	1st train-test	0.149	0.169	0.172	0.146	0.143
	2nd train-test	0.159	0.147	0.132	0.107	0.145
	Mean	0.154	0.158	0.152	0.127	0.144
NSE	1st train-test	0.954	0.902	0.907	0.931	0.953
	2nd train-test	0.942	0.946	0.93	0.973	0.942
	Mean	0.948	0.924	0.919	0.952	0.948

Table 9 Test statistics of the best OPELM, MARS, M5Tree, MARS-Kmeans and MLR in 12 h ahead streamflow prediction

Statistics	Dataset	OPELM	MARS	M5Tree	MARS-Kmeans	MLR
RMSE	1st train-test	0.872	1.076	1.091	0.882	3.097
	2nd train-test	0.643	0.645	0.787	0.391	2.437
	Mean	0.758	0.861	0.939	0.637	2.767
MAE	1st train-test	0.239	0.257	0.286	0.209	2.029
	2nd train-test	0.238	0.231	0.224	0.150	1.696
	Mean	0.239	0.244	0.255	0.180	1.863
NSE	1st train-test	0.877	0.813	0.808	0.875	−0.545
	2nd train-test	0.881	0.88	0.821	0.956	−0.714
	Mean	0.879	0.847	0.815	0.916	−0.630

regression (MLR). According to Table 7, for one hour ahead prediction, the five models showed relatively the same accuracy with negligible difference: (i) The mean MAE was ranged from $0.030 \text{ m}^3/\text{s}$ to $0.037 \text{ m}^3/\text{s}$ with no significant difference between the 1st and 2nd datasets; (ii) the models exhibited the same NSE values ($\text{NSE}=0.99$). However, when the models were compared taking into account the RMSE values (Table 7), estimates of hourly runoff showed a broad range of fluctuation and the obtained mean RMSE were ranged from $0.096 \text{ m}^3/\text{s}$ to $0.116 \text{ m}^3/\text{s}$. The lowest value was obtained using the MARS-Kmeans ($\text{RMSE}=0.096 \text{ m}^3/\text{s}$), with an improvement of 11.11%, 13.51%, 17.24%, 12.73% compared to OPELM, MARS, M5Tree, MLR models, respectively. Finally, for further comparison, results using the multi-model with simple averaging (MM-SA) were also provided in Table 7. It is clear that the MM-SA was more accurate than MARS, M5Tree and MLR models, worked equally with the OPELM model and was less accurate than the MARS-Kmeans with an increase in the RMSE value of about 11.11%. Table 8 shows the results of runoff forecasting for six hours ahead using the four machines learning models. The comparison between the models was reported in terms of mean RMSE, MAE and NSE values. In general, the agreement between measured and predicted hourly runoff is good: The NSE values are generally larger than 0.92 for most of the models, and the RMSE and MAE are reasonably low. The mean RMSE and MAE ranged from $0.478 \text{ m}^3/\text{s}$ to $0.627 \text{ m}^3/\text{s}$,

and from $0.127\text{ m}^3/\text{s}$ to $0.158\text{ m}^3/\text{s}$, respectively. For both models, the NSE values were varied between 0.919 and 0.952. The best accuracy was observed for MARS-Kmeans with NSE value higher by 0.42%, 2.94%, 3.47% and 0.42%, and with RMSE values lower by 2.64%, 21.12%, 23.76% and 3.63% compared to those of the OPELM, MARS, M5Tree and MLR models. Similarly, the MARS-Kmeans decreased the MAE of the OPELM, MARS, M5Tree and MLR by 17.53%, 19.62%, 16.44% and 11.81%, respectively. Nevertheless, the NSE values were the highest for one hour ahead and decreased with increasing horizon, from one to six hours in advance. For twelve hours ahead forecasting as shown in Table 9, the performances of the four models were dramatically decreased, and the four machines learning models exhibit a noticeable increase in the RMSE and MAE, and a significant decrease in terms of NSE values, compared to the one and six hours ahead horizons. The best accuracy was obtained using the MARS-Kmeans model with a mean NSE value higher than the values obtained using the OPELM, MARS, M5Tree and MLR by 4.04%, 7.53%, 11% and 168%, respectively. In addition, the MARS-Kmeans improves the RMSE and MAE of the OPELM, MARS, M5Tree and MLR models by 15.96%, 26.02%, 32.16%, 76.98% and by 24.68%, 26.23%, 29.41%, 168.78%, respectively. The OPELM model was ranked second, MARS in the third place and the M5Tree was the much weaker model with the lowest accuracy among the machine learning models.

Finally, we performed a global comparison between the EBA4SUB model and the developed four machines learning methods in forecasting hourly runoff. We have therefore tested the models for different recorded extreme runoffs, and in total, we have selected four events labeled as Event 1–Event 4. Results are depicted in Table 10 and Fig. 5. For Event 1, in general, it appeared that all machines learning models performed well, with NSE values of 0.994, 1.00, 0.993 and 0.993 for OPELM, MARS, M5Tree and MARS-Kmeans, respectively. The predicted hourly runoff is remarkably close to the corresponding in situ measured hourly runoff as shown in Fig. 5, where the data points were very less scattered. Conversely, the performances of the EBA4SUB model are significantly lower than those of the four machines learning models, and it is clear from Fig. 5, that the data were more scattered compared to the machines learning models. According to Table 10, the best accuracy was obtained using the OPELM model, having the lowest RMSE and MAE values. The OPELM model decreased the RMSE values by 9.77%, 5.42%, 4.84% and 81.55%, compared to the MARS, M5Tree, MARS-Kmeans

Table 10 Comparison of machine learning and EBA4SUB methods in modeling rainfall-runoff of various events

Method	RMSE	MAE	NSE	RMSE	MAE	NSE
	Event 1			Event 2		
OPELM	0.157	0.096	0.994	0.167	0.104	0.996
MARS	0.174	0.106	1.000	0.177	0.105	1.000
M5Tree	0.166	0.102	0.993	0.172	0.105	0.996
MARS-Kmeans	0.165	0.110	0.993	0.175	0.114	0.995
EBA4SUB	0.851	0.474	0.817	0.450	0.252	0.969
	Event 3			Event 4		
OPELM	0.196	0.107	0.970	0.170	0.086	0.998
MARS	0.217	0.115	1.000	0.171	0.088	1.000
M5Tree	0.170	0.104	0.978	0.171	0.101	0.997
MARS-Kmeans	0.167	0.117	0.978	0.169	0.102	0.998
EBA4SUB	0.397	0.165	0.877	1.038	0.553	0.908

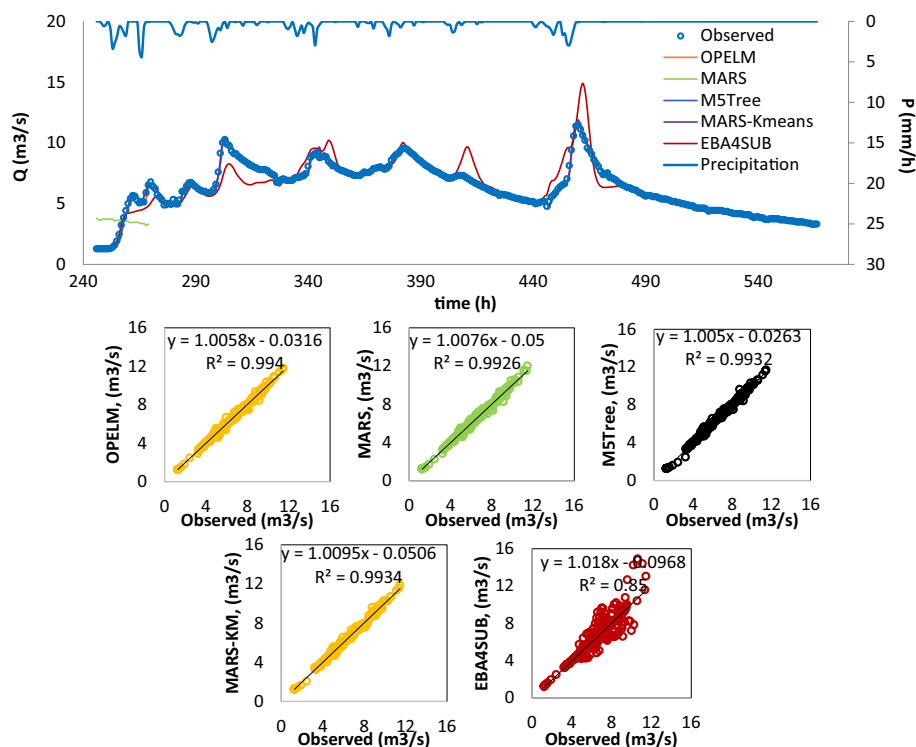


Fig. 5 Comparison of machine learning and EBA4SUB methods in modeling rainfall–runoff of event 1

and EBA4SUB, respectively. In addition, the OPELM model yielded a decrease in the MAE values by 9.43%, 5.88% and 12.72%, compared to the MARS, M5Tree and MARS-Kmeans, respectively, and dramatically compared to the EBA4SUB (79.74%). Regarding the Event 2, it is clear from Table 10 and Fig. 6, that MARS, M5Tree and MARS-Kmeans possess relatively the same accuracy with negligible difference in terms of RMSE, MAE and NSE values. The best accuracy was achieved using the OPELM model with the lowest RMSE and MAE values. For comparison, the RMSE of the OPELM was dropped from 0.177 to 0.167 m³/s (5.65% reduction), and from 0.172 to 0.167 m³/s (2.90% reduction), and from 0.175 to 0.167 m³/s (4.57% reduction), compared to MARS, M5Tree and MARS-Kmeans models, respectively. However, the difference between the EBA4SUB and the machines learning models was more obvious, and the difference was less significant compared to the Event 1. The NSE value is large for Event 2 ($NSE=0.969$) compared to the Event 1 ($NSE=0.817$). Furthermore, the difference in terms of RMSE and MAE between EBA4SUB and the OPELM for Event 2 is equal to 62.88% and 57.73%, respectively. Nonetheless, Fig. 6 demonstrates that EBA4SUB was able to predict the runoff data and reproduce high correlation between measured and predicted hourly runoff for Event 2, as the data points were very less scattered. For Event 3 as shown in Table 10, in overall, the predicted hourly runoff using the four machines learning models is all in good agreement with measured data, and none of the models has a clear superiority over the other, since the statistical indices fluctuated significantly from one model to another. Indeed, making direct reference to the

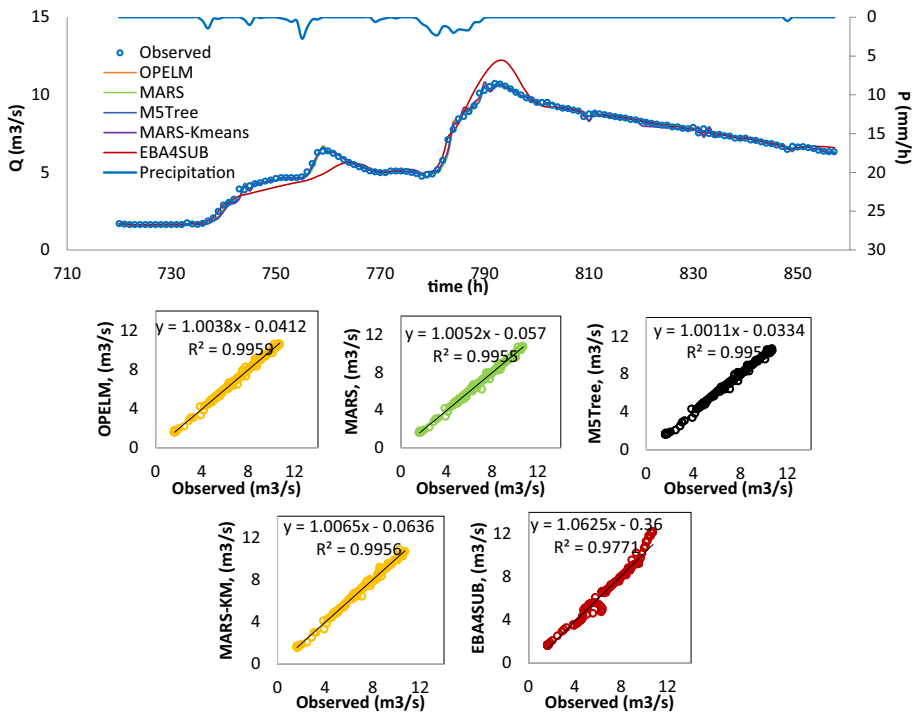


Fig. 6 Comparison of machine learning and EBA4SUB methods in modeling rainfall–runoff of event 2

RMSE value, MARS-Kmeans model was the best and exhibited the lowest RMSE value of $0.167 \text{ m}^3/\text{s}$. However, the lowest MAE value of $0.104 \text{ m}^3/\text{s}$ was obtained using the M5Tree significantly lower than the value obtained using the MARS-Kmeans (11.11% reduction). Eventually, as shown in Table 10, the MARS model exhibited a good prediction with NSE value of 1.00, more accurate than all the other models. Lastly, in comparison with the machines learning models, EBA4SUB has a considerably higher RMSE ($0.397 \text{ m}^3/\text{s}$) and MAE ($0.165 \text{ m}^3/\text{s}$) values and a relatively weaker NSE (0.877) value, which leads to a significantly less accuracy. This is not surprising since EBA4SUB has been designed to be a parsimonious model, in terms of input parameters, to be applied using the same information of the rational formula. Moreover, it should be emphasized, EBA4SUB estimates the runoff without any information on runoff data, i.e., it takes into account only rainfall data. One consequence of the previous statement on accuracy is certainly reflected in the scatterplot of Fig. 7, for which the data were more scattered and several outliers can also be easily individualized. Regarding the Event 4, the four machines learning models were significantly more accurate than the EBA4SUB, their superiority was quite obvious, and the later has achieved record levels regarding the RMSE and the MAE values of $1.038 \text{ m}^3/\text{s}$ and $0.553 \text{ m}^3/\text{s}$, greater than any values obtained for the first, second and third events. Between the machines learning models, the OPELM was the best accurate with the lowest MAE of $0.086 \text{ m}^3/\text{s}$, and with RMSE value of $0.170 \text{ m}^3/\text{s}$. Also, it is clear from the obtained results that the MAE largely fluctuated compared to the RMSE values, starting with the maximum values for MARS-Kmeans ($0.102 \text{ m}^3/\text{s}$), dropped to $0.101 \text{ m}^3/\text{s}$ and $0.088 \text{ m}^3/\text{s}$ for M5Tree and MARS

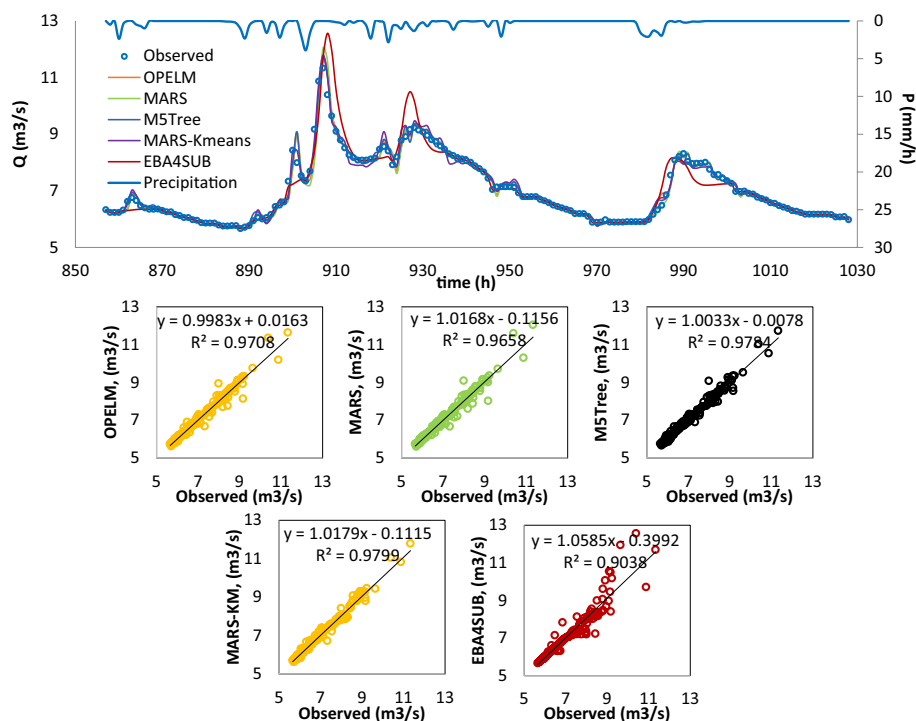


Fig. 7 Comparison of machine learning and EBA4SUB methods in modeling rainfall–runoff of event 3

models, respectively. The superiority of the machines learning models was clearly schematized in Fig. 8, for which the data points were less scattered for the machines learning compared to the EBA4SUB.

4 Conclusions

The study investigated ability of four machine learning (ML) methods, i.e., OPELM, MARS, M5Tree and MARS-Kmeans, in hourly rainfall–runoff modeling and compared their results with a conceptual event-based method, EBA4SUB, considering four observed events. The ML methods were also compared with the MLR method. Data were divided in two equal parts, and the developed models were tested using each part. A simple ensemble method was also applied to the results and compared with the single models. The following conclusions were reached after analysis of the outcomes produced by the employed methods.

- Excluding previous runoff information from the model inputs or including only rainfall information produces inaccurate models in hourly runoff prediction.
- Including previous runoff in the model's inputs considerably improves the accuracy (increments in RMSE and MAE are generally more than 90%).

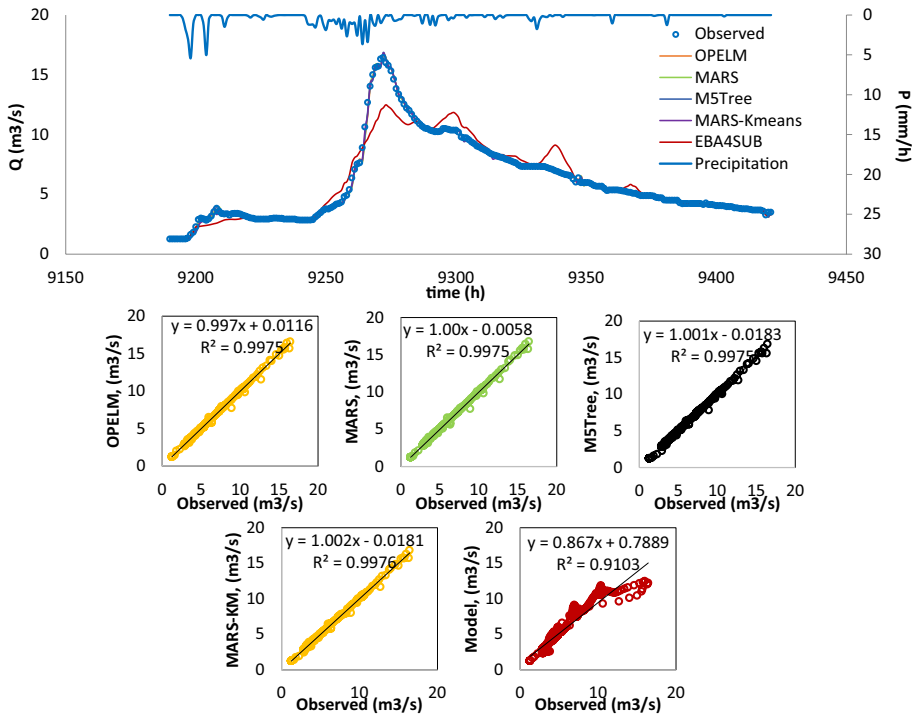


Fig. 8 Comparison of machine learning and EBA4SUB methods in modeling rainfall-runoff of event 4

- Comparison of the machine learning methods in modeling one hour ahead rainfall-runoff revealed that the MARS-Kmeans performed superior to the other alternatives. It is worth noting that this method is much more successful in case of only rainfall information is available compared to other three machine learning methods.
- Comparison of the methods considering four events showed that the all machine learning methods produced better accuracy than EBA4SUB. Among the machine learning methods, the OPELM provided slightly better accuracy than the others.
- Comparison of the methods in modeling runoffs in longer horizons (6- and 12-h) indicated that the MARS-Kmeans method outperformed the OPELM, MARS, M5Tree and MLR methods.

The main limitation of the presented study is the lack of data for a longer period. Hourly data of one catchment covering a 4-year period were used in the study, and for future studies, more data from different climatic regions should be considered. The proposed method could also be compared with some other ML methods such as neuro-fuzzy with meta-heuristic algorithms like gray wolf optimization, sine cosine algorithm, etc.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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